

VALENTENO LENORA

AI/ML ENGINEER | FULLSTACK DEVELOPER | DATA ENGINEER

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PROFILE SUMMARY

Currently developing AI systems at AiRentoSoft and WaveSkill, with expertise spanning Agentic Systems, Deep learning, Machine learning and scalable infrastructure. Dual-degree background in Computer Science (First Class Honours) and Information Systems provides both technical depth in AI/ML and Business systems perspective. Published a research paper and Open Source package for community. Passionate about transforming complex AI innovations into deployed solutions that deliver measurable real-world impact.

WORK EXPERIENCE

AI/ML Engineer - Freelance - WaveSkill Pvt. Ltd.

2025 (December) - Present

- Designed and engineered AI-powered financial document extraction pipelines integrating hybrid OCR and multimodal LLMs to parse tabular data and financial statements.
- Built the fundamental analysis and metrics calculation engine for Colombo Stock Exchange (CSE) filings, generating automated corporate valuation insights.
- Architected an agentic e-commerce conversational assistant for TOGO with multi-agent orchestration (CrewAI, LangGraph) and observability via Langfuse.
- Created a custom phonetic dataset from scratch and trained a research-grade Coqui VITS Text-to-Speech (TTS) model for the Sinhala language.

AI/ML Engineer - Intern - AiRentoSoft Pvt. Ltd.

2026 (March) - 2026 (September) | 6 months

- Engineered an automated Vehicle Damage Detection system combining fine-tuned YOLO computer vision models with FastAPI microservices and mobile inspection app integration.
- Collaborated on core features across the AiRentoSoft ecosystem (React web portal, reservations plugin, .NET backend, and v2.0 mobile app released on App Store & Play Store).
- Designed custom OCR document scanning and credit card extraction modules to streamline customer onboarding and verification workflows.
- Implemented an automated B2B lead generation pipeline utilizing Playwright, Selenium, and custom ETL workflows to process targeted car rental business prospects.
- Managed sprint deliverables, code reviews, and cross-functional feature planning using Jira and Azure DevOps.

AI/ML Engineer - Intern - Residue Solutions Pvt. Ltd.

2025 (September) - 2026 (February) | 6 months

- Technical Lead for a 5-member AI engineering intern team, driving system architecture, agile sprint planning, code reviews, and delivery milestones.
- Architected FashionHub.ai, an intelligent shopping platform featuring ReAct DAG-based agent execution, real-time Weaviate vector retrieval, and Kafka event streaming.
- Integrated a modular multi-LLM factory (Gemini 1.5, Llama 3 via Ollama) and Google Vertex AI Virtual Try-On pipelines for context-aware customer personalization.

System Support Officer - IT - Mercantile Investments FPLC

2022 | 3 months

- Executed enterprise hardware diagnostics, LAN/WAN maintenance, and component-level repairs across distributed corporate workstation infrastructures.
- Delivered Tier-1/Tier-2 Help Desk technical support, rapidly diagnosing OS, software, networking, and peripheral issues to ensure seamless business continuity.

EDUCATION

BSc (Hons) Computer Science (First Class Honours)

2022 - 2025

University of Plymouth, UK (NSBM Green University)

BComp (Hons) Information Systems (4th Year)

2023 - Present

University of Sri Jayewardenepura

Dip. in Technology

2019

Esoft Metro Campus

A/L - Math Stream (A1, C2 (O/L: A7, B2))

2008 - 2021

St. Peter's College, Colombo - 4

PROJECTS

Work Projects

Vehicle Damage Detection System

YOLO, PyTorch, FastAPI, React, HuggingFace, WandB, Kaggle, MLOps, Roboflow

Developed a Vehicle Damage Detection system consisting of a combination of YOLO models with a FastAPI wrapper hosted in Huggingface spaces. Pre-trained the models on Image datasets, versioned and deployed using MLOps tools. Also further developed the live mobile app to support new features which is in Playstore and Appstore.

AiRentoSoft System

React, Node.js, .NET, Retel.ai, OCR, Azure DevOps

Collaborated in Developing and Implementing new features including AI Integrations. Worked on the React Web application, Reservations Plugin, .NET API, and developed the 2.0 version of the Mobile app with similar features to the main system and released documentation in AZURE DevOps. Added AI Integrated features like document scanning, card scanning using OCR models in both web and mobile apps. Also managed and worked on the AI Caller agent in Retel.ai for client requirements.

Business Leads Extraction

Playwright, Selenium, Python, ETL

Implemented a business leads extraction pipeline for extracting potential clients in USA based car rental businesses using Automation scripts and web scraping and using ETL pipelines to process data into understandable formats for business support division to use.

Sinhala TTS Model

Coqui VITS, PyTorch, Kaggle

Developed a research purpose private TTS model with support for Sinhala language. Created a private dataset from scratch for VITS format to train the Speech Model. Trained on Kaggle dual T4 GPUs and gained research level quality output for product pitching and further development with funds.

TOGO - AI Shopping Assistant

FastAPI, Postgres, CrewAI, LangGraph, Langfuse

Developed an Agentic Shopping assistant for e-commerce website for baby care products. Implemented features in Product search, QnA using store policies and customer support. This is live on the TOGO website.

FashionHub.ai

Next.js, FastAPI, Weaviate, Ollama, LangGraph, LangChain, Kafka, Groq, Gemini, MongoDB, Google Vertex AI

Developed an intelligent AI shopping assistant that revolutionises the retail experience by utilising a ReAct capable Agent with DAG-based task execution. The platform features a high-throughput MongoDB to Weaviate Kafka pipeline for real-time data synchronization. Built on a scalable FastAPI microservices infrastructure, the system integrates a modular multi-LLM factory (Gemini, Llama) and Google Vertex AI for generative Virtual Try-On capabilities, delivering a seamless and context-aware customer assistance. Used LangChain & LangGraph for LLM orchestration and agent state management.

Stock Market API

FastAPI, LlamaIndex, OpenCV, Groq, MongoDB, LLMs, OCR

Designed and engineered a scalable financial intelligence API using FastAPI for Fundamental analysis of Colombo Stock Exchange (CSE) filings. Built the analysis engine that can calculate core financial ratios and growth metrics, integrating AI-driven insights to provide comprehensive qualitative and quantitative evaluations of corporate performance. Additionally, developed an extraction engine that utilizes hybrid OCR and semantic mapping with Large Language Models (LLMs) to transform unstructured quarterly statements into standardized datasets.

Personal Projects

Accordo.ai - Music Analysis Platform

TensorFlow, RNN, Bi-LSTM, Flutter, FastAPI, Celery, Redis, MongoDB, TensorFlow Serving, GCP, MinIO S3

Developed an intelligent system that performs musical analysis on audio for musicians. The project explored advanced deep learning techniques, including Bi-LSTM networks and time-frequency feature extraction, trained on real-world music datasets and achieved +80% accuracy. Focused on research and development of robust music information retrieval (MIR) model, with a practical Implementation using FastAPI, MongoDB, and TensorFlow Serving with a Flutter-based mobile UI for real-time analysis and playback integration. Deployed using a CI/CD pipeline to GCP.

Univize - University Social App

Django, FastAPI, Next.js, Neo4j, PostgreSQL (Supabase), MongoDB, Redis, Supabase, Azure, GitHub Actions, Azure Bicep

Developed an Social application for university students with features including, medial sharing, direct messaging and forum based discussions. Led the development team in system design to development, testing and deployment. Used Azure as the hosting platform with services like container registry, secret management and containerized apps. Developed the main backend in Django and FastAPI with PostgreSQL (Supabase), MongoDB, Neo4J, Redis and Supabase. Built the CI/CD pipeline with GitHub Actions and Azure Bicep scripts for infrastructure provisioning and Integration. This was done as the Group project for USJP.

Real-Time Fraud Prevention & Intelligent Analytics Engine

Apache Spark 3.5, Apache Kafka, Apache NiFi, MLflow, LightGBM, Prometheus, Grafana, Google Cloud BigQuery, PySpark

Architected an enterprise distributed streaming intelligence pipeline to detect fraudulent transactions in real time (<50ms SLA). Pruned 339 proprietary V-features from 590k+ IEEE-CIS records to engineer a 30-feature reproducible pipeline. Formulated financial cost-matrix threshold optimization ($\tau^* = 0.10$) to minimize chargeback losses vs false friction. Deployed vectorized PyArrow/Pandas UDF scoring on Spark Structured Streaming with dual-sink routing to Kafka alerts and GCP BigQuery, instrumented with Prometheus metrics and a pre-configured Grafana telemetry dashboard.

EduAgent - AI-powered Research Assistant

CrewAI, LangChain, Ollama, Mistral LLM, ChromaDB, HuggingFace

Engineered an autonomous multi-agent research assistant using CrewAI, LangChain, and Mistral LLM (via Ollama) for academic paper ingestion, synthesis, and Q&A. Implemented hybrid semantic retrieval with ChromaDB vector store and HuggingFace embeddings, automated paper section summarization, topic extraction, and citation-grounded conversational search.

AI-Powered Recruitment Platform

Streamlit, LangChain, Groq, Qdrant, HuggingFace, MongoDB, Llama LLMs, RAG

Built an AI-powered talent acquisition platform streamlining candidate screening via semantic resume-to-job matching with RAG architecture. Utilized Streamlit, Qdrant vector database, MongoDB, and Groq/Llama LLMs to parse multilingual resumes, generate tailored JD requirements, compute similarity rankings, and accelerate candidate shortlisting.

Vision Reasoning Surveillance System

PyTorch, OpenCV, YOLO, LangChain, Ollama, LLM, Streamlit, Telegram Bot

Engineered an end-to-end intelligent video surveillance and threat detection system combining real-time YOLO computer vision with Multimodal LLM reasoning. Automated live RTSP stream object detection, temporal anomaly localization, and structured scene analysis, dispatching instantaneous contextual alerts via Telegram Bot API.

Brain Tumour Detection Model

PyTorch, TensorFlow, OpenCV, U-Net, CNN, Alumentations, Kaggle

Developed high-precision medical imaging segmentation models using U-Net encoder-decoder architectures in PyTorch and TensorFlow across 2D/3D MRI Kaggle datasets. Applied Alumentations pipelines for data augmentation, optimized composite DICE/IoU loss functions, and achieved ~90% segmentation accuracy with volumetric lesion quantification.

Weekly Report Generator & Team Dashboard

FastAPI, MongoDB, React 19, TypeScript, Zustand 5, Tailwind CSS, LangChain, LangGraph, Groq, Docker

Engineered an asynchronous Weekly Report Generator and Team Dashboard with RBAC (Contributors, Managers, Admins), ISO calendar week management, immutable version snapshots, and live KPI analytics. Built an AI Assistant using LangGraph StateGraph, LangChain RAG, and Groq LLM with a 5-response rolling summarizer and session memory. Containerized full stack with Docker Compose.

Data Analytics Dashboard

Python, SQL, Snowflake, ETL, XGBoost, Streamlit

Built an end-to-end real-time data engineering and intelligence pipeline extracting LinkedIn job market datasets via Python ETL into Snowflake cloud data warehouse. Trained XGBoost regression and classification models to identify emerging skill demand trends and salary benchmarks, visualizing live predictive insights on an interactive Streamlit dashboard.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#, Java, C, PHP, Dart, Bash

AI / ML & LLMs: TensorFlow, PyTorch, Transformers, LangChain, LangGraph, CrewAI, LlamaIndex, Scikit-Learn, LightGBM, XGBoost, Groq, Gemini, HuggingFace, OpenCV, Ollama, Unsloth, Coqui VITS, Retel.ai, Langfuse

Full Stack: FastAPI, React 19, Next.js, Node.js, TypeScript, Zustand 5, Tailwind CSS, Django, Streamlit, React Expo, Flutter, .NET, RESTful APIs

Distributed & Data: Apache Spark 3.5, Apache Kafka, Apache NiFi, Google Cloud BigQuery, Snowflake, PySpark, PyArrow, Real-time Streaming ETL

DevOps & MLOps: Docker, Docker Compose, MLflow, Prometheus, Grafana, GitHub Actions, WandB, Ansible, Terraform, Azure Bicep, Azure DevOps, CI/CD

Databases & Vector: MongoDB, PostgreSQL, MySQL, Supabase, BigQuery, ChromaDB, Qdrant, Weaviate, Neo4j, PGvector, Redis, MinIO S3

Cloud Platforms: Google Cloud Platform (GCP), Microsoft Azure, Streamlit Cloud, Google Vertex AI, HuggingFace Spaces, Kaggle, Roboflow

Soft Skills: Technical Leadership, Problem Solving, Agile / Scrum, Cross-functional Collaboration, System Design, Time Management

PUBLICATIONS & CONTRIBUTIONS

Musical Chord Recognition using Deep Learning

Research paper published in the journal at the NSRSIT'25 Conference.

accordo.ai

Published PYPI package with the developed DL Models for community use.

ACTIVITIES & EXTRACURRICULARS

- Bassist at In the Zone Band (2025 - Present), Unstoppable Band (2023 - 2025)
- Choir / Music Director at Music Club of NSBM (2023 - 2024)
- Member of the ICT Club, Maths & Science Society of St. Peter's College (2019 - 2021)
- Keyboardist / Choir Leader at Our Lady of Lourdes Choir
- Member of the College Under 13 Div. 1 & 3 Cricket team (2013 - 2014)
- President of the YCS Movement (2020 - 2021) & DH. Prefect of Our Lady of Lourdes Church (2017 - 2018)

REFERENCES

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